

From Enterprise Knowledge Graphs to Policy-Compliant Agents

Governing Data, Models, and Autonomous Curation

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Overview

SECTION 1 · 2 MIN

Introduction

why governance changed; governance of the graph vs in the graph; four contributions

SECTION 2 · 1 MIN

Related work

what ODRL, DPV, SHACL, AIRO and policy engines already govern, and what agents change

SECTION 3 · 3 MIN

The Metagraph

the system: one statement, one gateway, one log, one loop

SECTION 4 · 6 MIN

Governance as graph content

the four mechanisms · regulatory model · agentic curation · model governance · the worked example

SECTION 5 · 1 MIN

Alignment with standards

where ODRL, DPV and SHACL reach, and where they stop; the mapping itself is additional material

SECTION 6 · 3 MIN

Discussion

in-graph governance; what changes as data; limits and next steps

The thread: one policy, MAS Notice 626 § 8.2, followed from regulatory text to an approved write and its audit entry.

Agentic knowledge graphs



The graph is no longer only an asset to be protected. It is the medium through which autonomous systems decide what to do next.

Governance of the graph, and governance in the graph

GOVERNANCE OF THE GRAPH · THE REQUIREMENT

control and account for how the graph changes and how it is used

IN-GRAPH GOVERNANCE · THE DESIGN POSITION

the rules themselves are graph content, in the same store as the knowledge they govern

regulatory obligations · model documentation · the curators' behavioural policies

FOUR CONTRIBUTIONS

§4.1

regulatory knowledge model

regulations as traversable obligation → control → risk-factor chains, connected to the detection logic

§4.2

model governance layer

the artifacts SR 11-7, SS1/23 and the EU AI Act require, as entities linked to the components they document

§4.3

policy-compliant agents

curators operate under behavioural policies stored in the graph they curate, with write modes, shapes and an attributed log

§5

alignment analysis

the mechanisms mapped onto ODRL, DPV and SHACL; what current standards do not cover

Mature standards cover pieces of this. Prescriptive policy for curators, graduated trust and per-mutation audit have had much less attention.

What the standards already govern

ACCESS AND USAGE
CONTROL

ODRL 2.2

domain-agnostic; profiles add domain concepts; enforceable with ODRE, FORCE; formal semantics beyond permitted/prohibited; used in Gaia-X and IDS

PRIVACY, DATA
PROTECTION, AI METADATA

DPV 2.0

jurisdiction-agnostic core; GDPR, AI Act and DGA in extensions

STRUCTURE · AI RISK

SHACL · AIRO

SHACL validates graph structure; AIRO maps the EU AI Act into an ontology for risk modelling

POLICY EVALUATION

XACML · OPA

attribute-based access control; infrastructure policy engines beside the data

DOCUMENTATION

model cards · dataset cards · capability cards

what a model, a dataset or an agent's capabilities are for, how they were validated, where they stop -- the artifacts the regimes effectively mandate

WHAT AGENT PARTICIPATION CHANGES -- IN KIND, NOT DEGREE

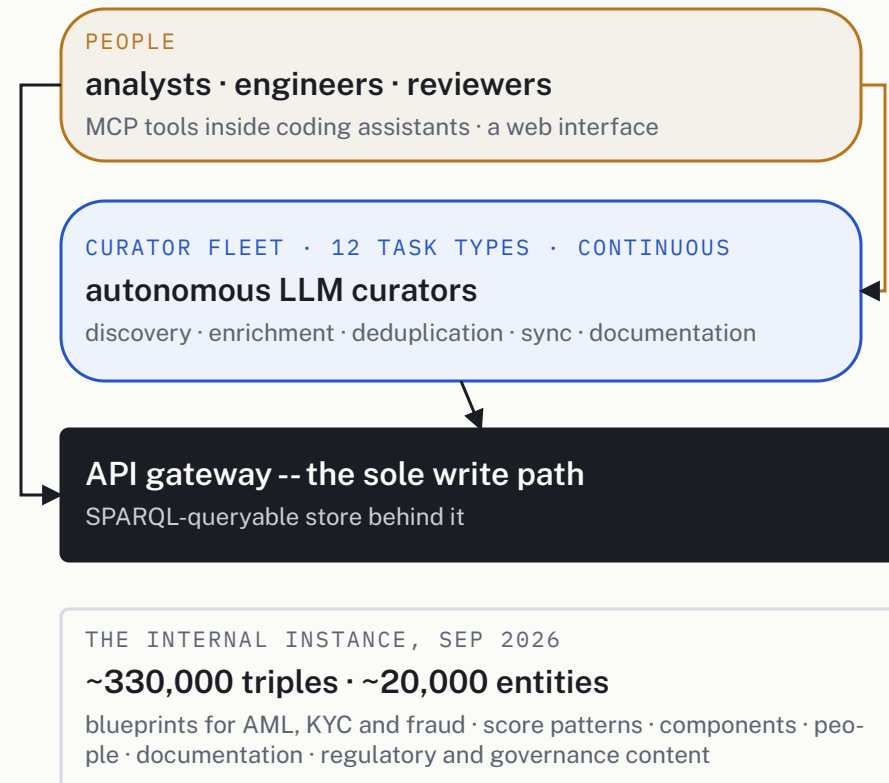
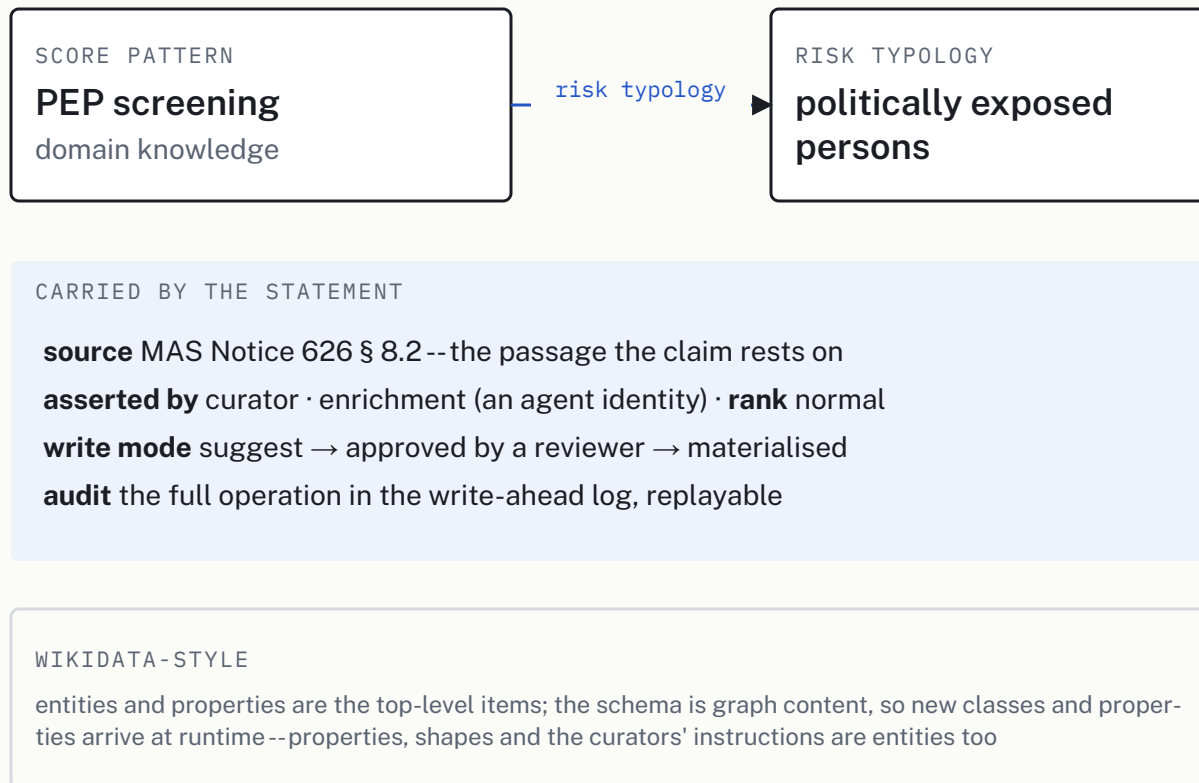
A policy unambiguous to a compliance officer may be ambiguous under an LLM's interpretation.

So behavioural rules must be explicit, attributed, validated structurally where possible, and audited per mutation.

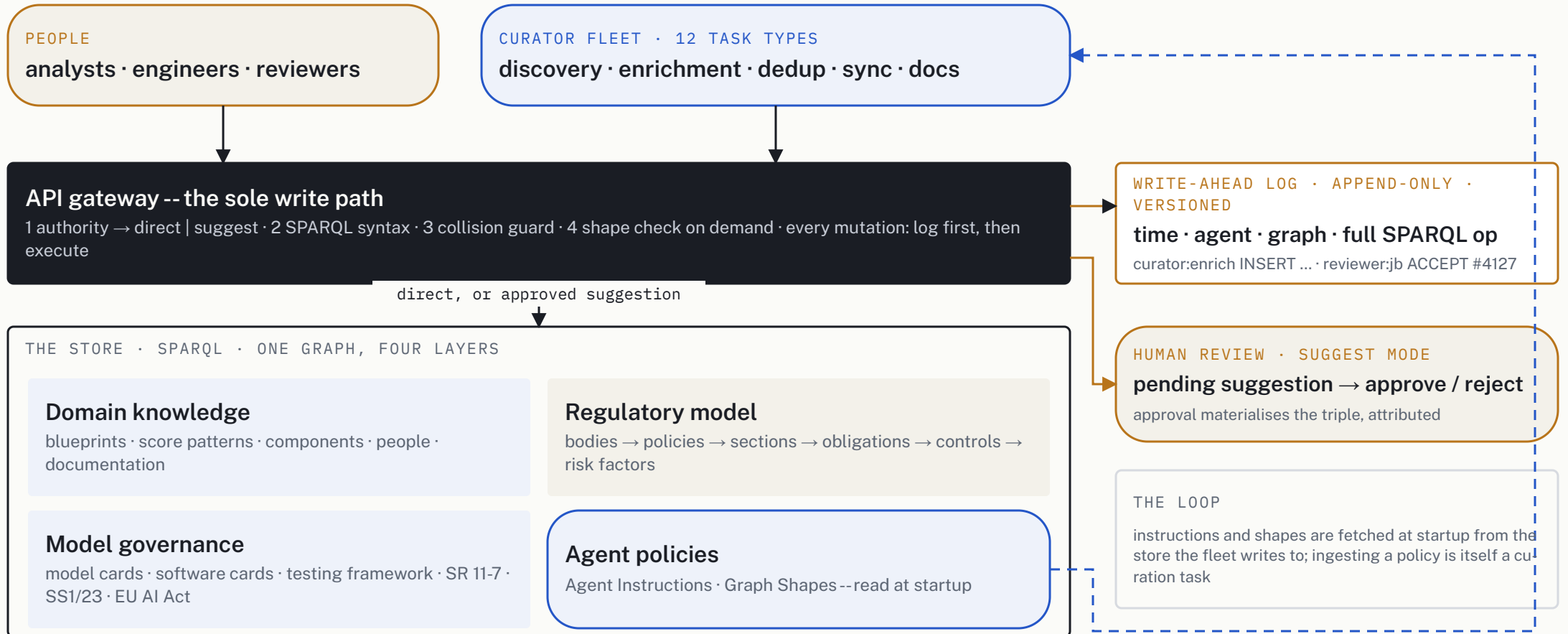
And the graph's semantics are partly emergent -- agents extend the schema as data arrives -- so the governance layer must evolve with it.

These standards govern data: who may access, process or share it. The combination an agent-curated graph needs is the gap this paper works in.

The Metagraph



The governance architecture



Four things the word “policy” names here

GOVERNS THE DOMAIN

SOURCE DOCUMENT

MAS Notice 626 · § 8.2

the passage a curated interpretation rests on
ingested by a curator as an entity with 37 sections;
17 bodies · 9 jurisdictions · 15 policies

entity · regulatory layer

OBLIGATION · CONTROLS · RISK FACTORS

apply enhanced due diligence to politically exposed persons (PEPs)

→ PEP screening · source-of-wealth · senior-management approval
the domain requirement, traversable to the detection logic
12 obligations · 8 controls · 8 risk factors

entities · imposes · requires · identifies · risk level

GOVERNS THE CURATOR

AGENT INSTRUCTION

*“Verify claims against an authoritative
source before writing.”*

priority critical · categorised · targets curator types
read by SPARQL at startup into the system prompt

entity · agent-policy layer · behavioural

TEXT, INTERPRETED

GRAPH SHAPE

score pattern

required typology · entity types · traversal · blueprints
banned the generic “related to” property
compliance computed per entity, on demand, by an endpoint

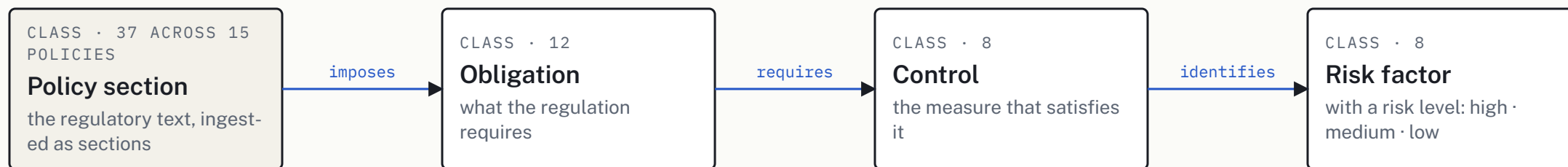
entity · agent-policy layer · structural

STRUCTURE, CHECKED

Four mechanisms govern the curators

MECHANISM	WHAT IT DECIDES	WHAT IT CANNOT SEE	ENFORCED BY
Write modes direct · suggest, by capability tier	whether an assertion lands now or waits as a pending suggestion for a person to approve	whether the assertion is true -- a tier says how much an agent may do, not whether it is right	gateway
Agent Instructions prioritised natural language, in the system prompt	what a curator verifies, which properties it prefers, how it resolves choices no shape can	its own compliance: an LLM interprets the text, so following it is probabilistic	the model
Graph Shapes required · optional · banned properties per class	whether an entity has the structure its class demands; compliance per entity, on demand	a well-formed entity carrying a wrong claim -- structure, not meaning	gateway · API
Audit + graduated trust attributed write-ahead log · capability tiers · write targets	who changed what, when, under which operation; how much an agent may do -- deny · read-only · propose · act -- and where a write lands: shared · private · draft · suggest	nothing in advance: a logged action can still be a mistake, the log makes it findable; the authority score ranks search results, it is not trust	gateway

The regulatory knowledge model



WHAT THE EDGES LET US ASK, IN EITHER DIRECTION

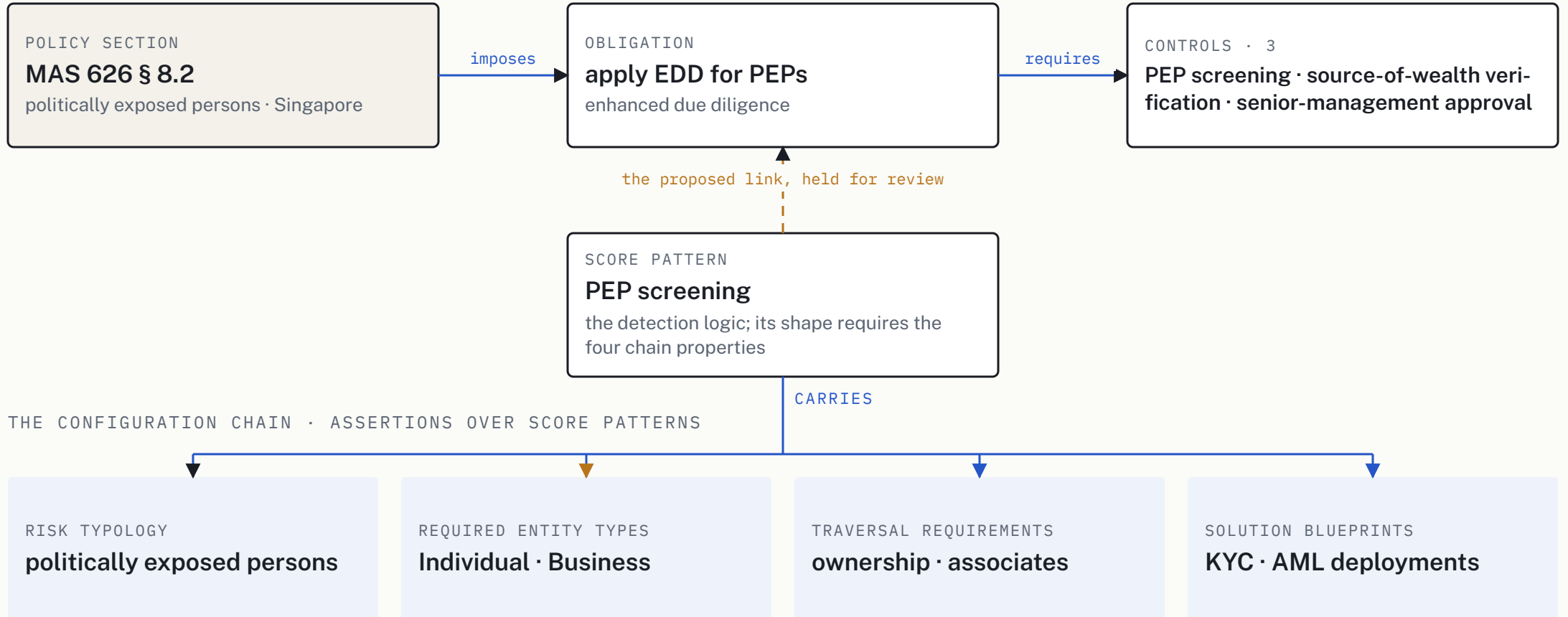
→ “What controls implement this obligation?” ← “Which obligations does this control satisfy?”

→ “Which risk factors does this section address, at what level?” ← “Which sections, across jurisdictions, reach this risk factor?”

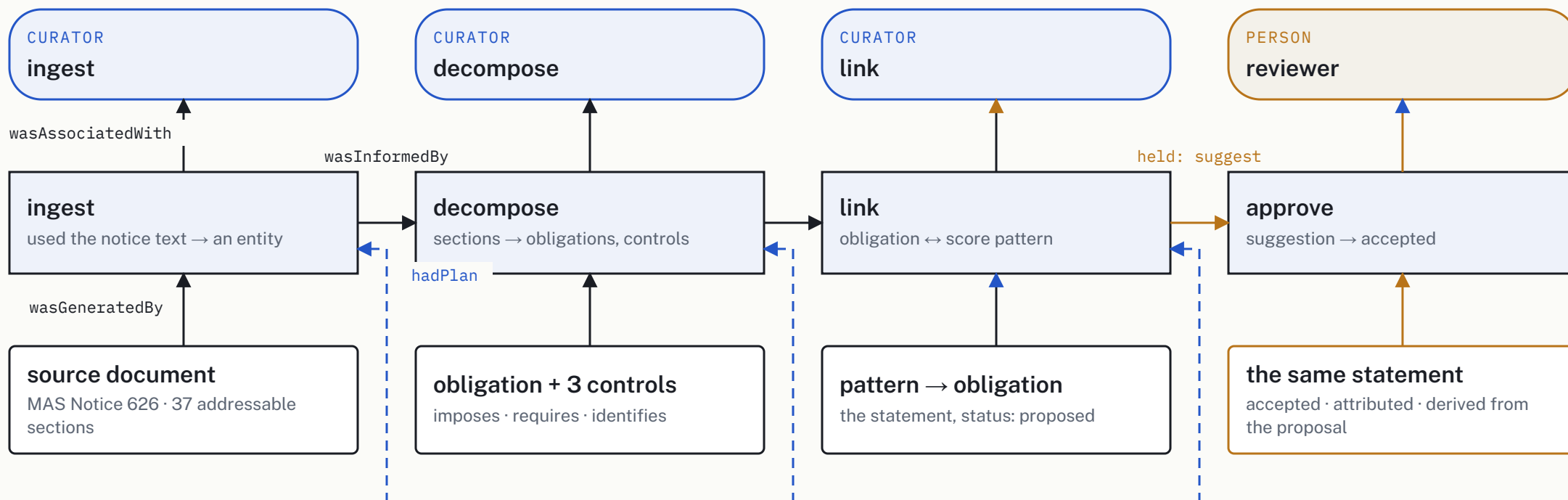
17 regulatory bodies · 9 jurisdictions · FATF, MAS, FCA, EBA, FinCEN and others

These edges record a curated interpretation that a reviewer can inspect against its source -- a query result, not a document exercise.

MAS Notice 626, decomposed and connected



Curation is agentic end to end

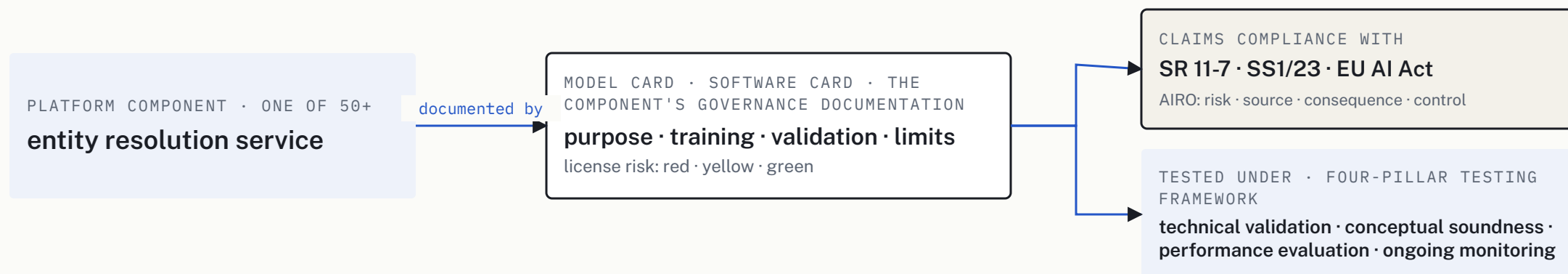


AGENT INSTRUCTIONS + GRAPH SHAPES · PROV:PLAN OF EVERY CURATOR ACTIVITY

“verify against an authoritative source before writing” · the score-pattern shape

read from the store at startup; the dashed edges are hadPlan; the chain is the paper's MAS 626 example, kept illustrative

Model governance as graph content



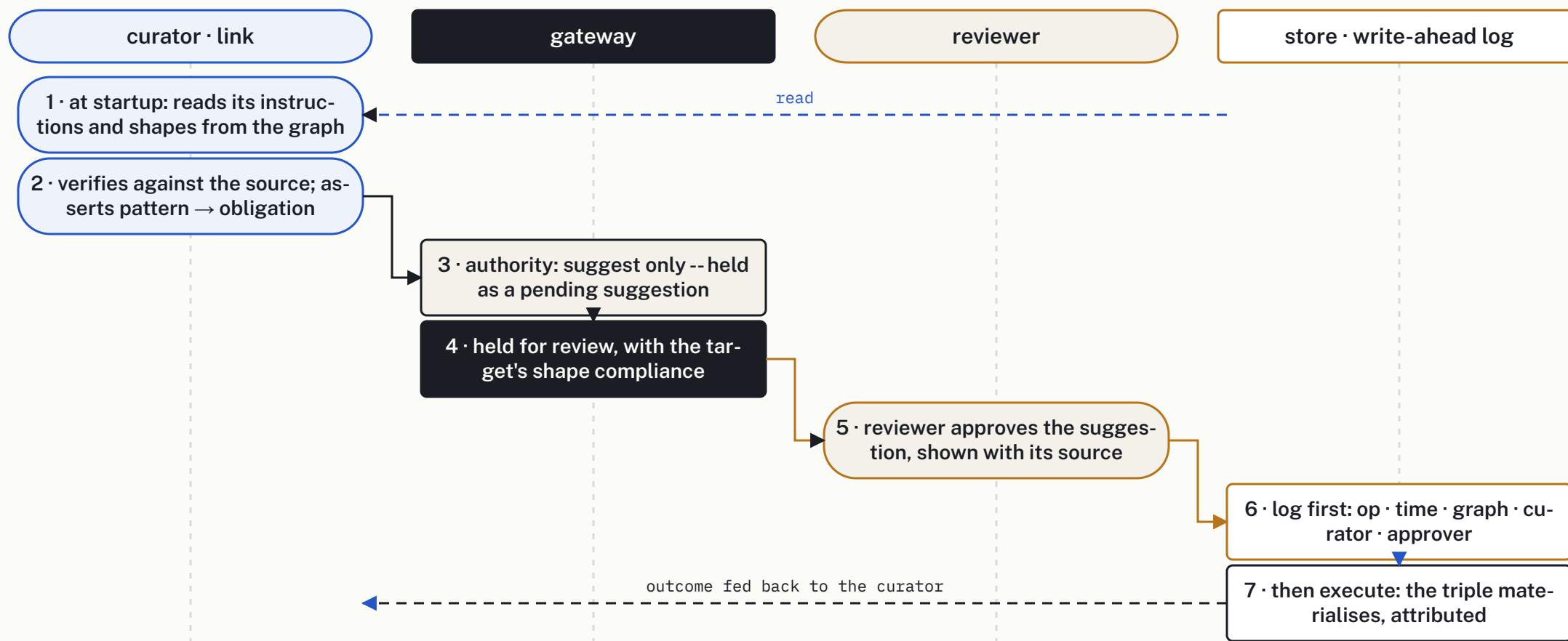
A GAP IS A QUERY, NOT AN AUDIT

```
SELECT ?c WHERE { ?c a mg:Component . FILTER NOT EXISTS { ?c mg:documentedBy ?card } }
```

“Which components lack model cards?” · “Which cards claim SR 11-7 compliance?” · “How complete is each client deployment's governance set?”

A 31-entity model-risk subgraph; governance maturity queryable across the portfolio -- global custody and retail banks, a securities-market utility, professional-services firms. Coverage targets review; it does not establish compliance by itself.

§4 · GOVERNANCE AS GRAPH CONTENT

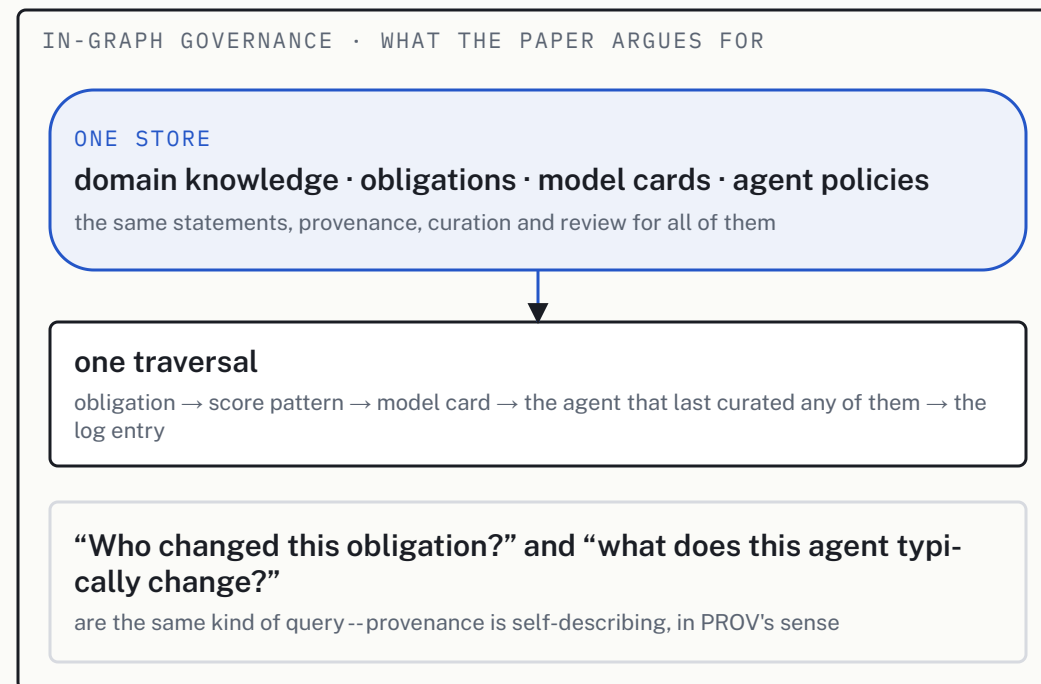
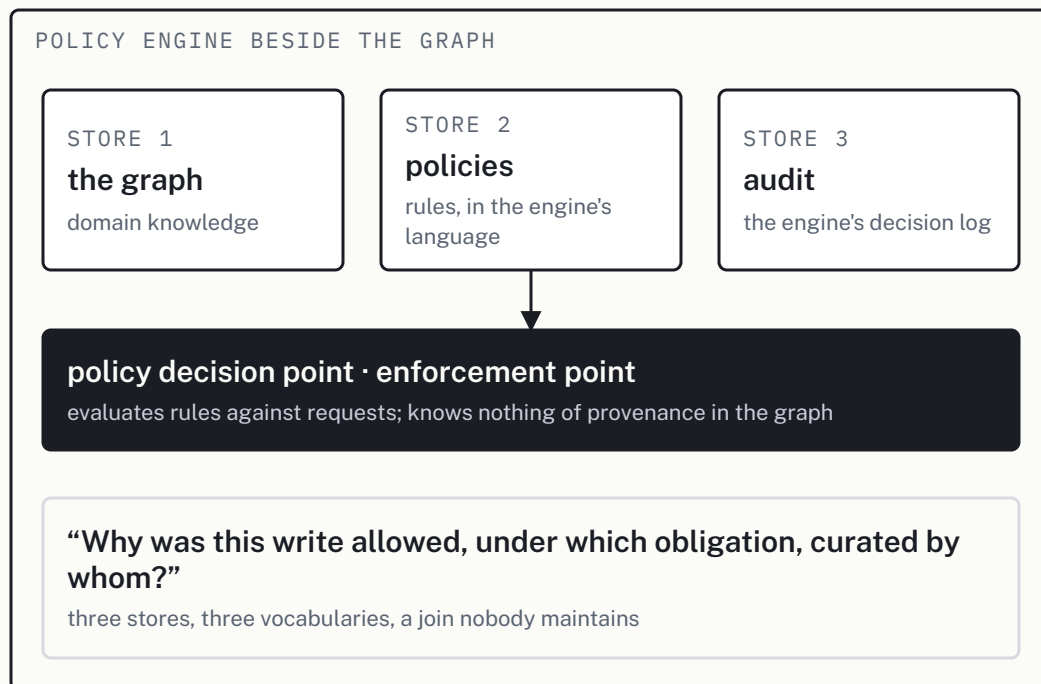


Where the standards reach, and where they stop

REQUIREMENT OF AN AGENT-CURATED GRAPH	ODRL	DPV	SHACL	METAGRAPH MECHANISM
Data access control	●	○	—	write modes
Privacy metadata	—	●	—	attribution properties
Structural validation	—	—	●	Graph Shapes
Agent behavioural governance	○	—	—	Agent Instructions
Obligation traceability	—	○	—	obligation → control chains
Model governance artifacts	—	○	—	governance-documentation links + framework entities
Per-mutation audit trail	—	—	—	agent-attributed write-ahead log
Graduated trust	—	—	—	capability tiers · write targets
Review workflow	○	—	—	suggestion pipeline

● covered · ○ partial · — out of scope · shaded: no current standard covers it

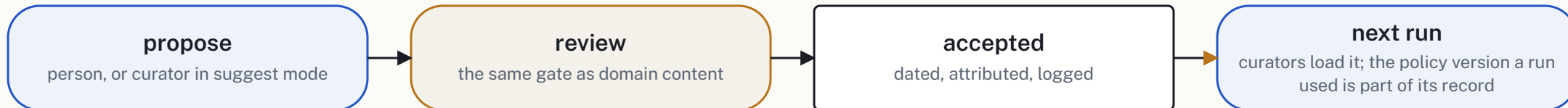
In-graph governance



Governance content is maintained by the same curation machinery as domain content: a new policy is ingested as content, and agents adapt on the fly.
Discussion, not disqualification: a policy engine beside the graph is the right tool where the rules are data-access rules alone.

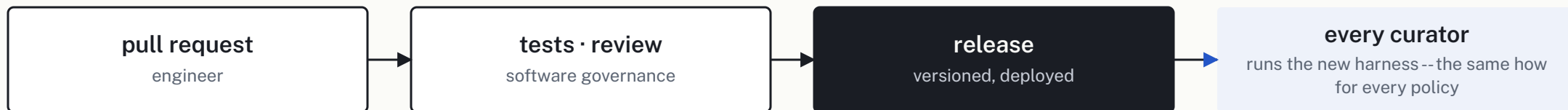
Policy change and software change

CHANGE THE DATA · CURATION, UNDER REVIEW, ATTRIBUTED



revise an instruction's text, priority or scope · add an obligation and its control mapping · tighten a shape with constraints it already supports · re-target an instruction at one curator type

CHANGE THE SOFTWARE · A RELEASE, WITH ITS OWN GOVERNANCE



add a tool a curator can call · add a hard enforcement check at the gateway · implement a new policy evaluator · give the harness a capability it lacks

Ask of any proposed change: does it use representations and tools the harness already has? Then it is data.

Limitations and next steps

THE LIMITATION

Agent Instructions are natural language interpreted by LLMs, not rules executed by a reasoner.

compliance is probabilistic -- a policy unambiguous to its author can admit several machine interpretations

MITIGATION

hard enforcement in code where a rule is expressible in code; shapes where it is structural

the log and suggest mode for the rest

NEXT STEPS

ODRL profile

write-mode policy with conflict detection over the formal semantics

DPV 2.0 mapping

the model-governance layer on the AI Act extension, for interoperable reporting

SHACL export

Graph Shapes as standard SHACL; the in-graph form stays authoritative

formalised instructions

enough to detect contradictions before agents meet them

roles, federation, multitenancy

explicit roles for humans and agents across deployments

close the loop

one agent follows a policy; others evaluate, score and curate better policy-following

Questions



Backstory

FIRST WEEK OF AUTONOMY

concurrent curators made conflicting edits nobody could assign

MULTI-USER DEPLOYMENT

more than one organisation's curators writing to one store

RECURRING SEMANTIC ERRORS

generic properties, duplicates, inconsistent dates --structure alone could not prevent them

THE STRUCTURAL REMAINDER

what natural-language rules fixed only probabilistically

THE ACTIVATION GAP · 2026

13 of 15 instructions in the graph governed nothing: the reader filtered on a text prefix, not on acceptance

Attribution

agent identity on every write

Write modes

direct · suggest

Agent Instructions

behavioural policy as data

Graph Shapes

structural policy as data

Activation as an edge

a policy is active when attached ('governed by') and the reader filters on that

Each control answered a failure in operation, then became data in the graph, not code. The policy layer records what broke, in order.

Metagraph → Semantic Web mapping

Additional material, after questions. A provisional mapping: what runs today is the in-graph representation; the standards forms are where we intend to meet.

ODRL 2.2 · WRITE POLICY AS A PROFILE

planned (paper §5)

```
mg:writePolicy a odrl:Set ;
  # direct write: a permission with an authority constraint
  odrl:permission [ odrl:action mg:write ; odrl:assignee mg:Curator ;
    odrl:constraint [ odrl:leftOperand mg:authority ;
      odrl:operator odrl:gteq ; odrl:rightOperand 0.8 ] ] ;
  # suggest mode: a permission carrying an approval duty
  odrl:permission [ odrl:action mg:write ; odrl:assignee mg:Curator ;
    odrl:duty [ odrl:action mg:approve ; odrl:assignee mg:Reviewer ] ] ;
  # a banned property: a prohibition
  odrl:prohibition [ odrl:action mg:assert ; odrl:target mg:P11 ] .
```

Evaluable with the ODRL formal semantics and the interoperable evaluators (Slabbinck et al., ESWC 2025); interoperable with Gaia-X and IDS policy frameworks.

DPV 2.0 · AI EXTENSION

aligned (paper §5)

```
mg:erService a ai:AISystem ;
  dpv:hasRisk mg:missedPepRisk ;
  dpv:hasTechnicalOrganisationalMeasure mg:erModelCard .
mg:missedPepRisk a ai:Risk ;
  dpv:isMitigatedByMeasure mg:ongoingMonitoring .
mg:ongoingMonitoring a ai:Measure .    # one of the four pillars
AI system · risk · measure, with AIRO's risk source and consequence as alignment
points; the EU AI Act, SR 11-7 and SS1/23 as entities the cards point at. Difference:
we run these as live queries, not static annotation.
```

SHACL · GRAPH SHAPES

export planned (paper §6)

```
mg:ScorePatternShape a sh:NodeShape ;
  sh:targetClass mg:ScorePattern ;
  sh:property [ sh:path mg:P468 ; sh:minCount 1 ] ;    # required
  sh:property [ sh:path mg:P469 ; sh:minCount 1 ] ;
  sh:property [ sh:path mg:P11 ; sh:maxCount 0 ] .    # banned
```

Functionally SHACL. Two deliberate departures: the shape is a first-class entity under the same curation and provenance as domain content, and compliance is computed per entity on demand through an API rather than in batch validation.

PROV-O · THE LOG AND ITS PLAN

proposed here

```
mg:run42 a prov:Activity ; prov:used mg:mas626_s8_2 ;
  prov:qualifiedAssociation [ prov:agent mg:curatorLink ;
    prov:hadPlan mg:instr_verifyFirst ] .    # the Agent Instruction
mg:stmt17 a prov:Entity ; prov:wasGeneratedBy mg:run42 ;
  prov:wasAttributedTo mg:curatorLink .
mg:approval9 a prov:Activity ; prov:wasInformedBy mg:run42 ;
  prov:wasAssociatedWith mg:reviewerJB .
```

Every log entry is an Activity; the curator a SoftwareAgent; the instruction a Plan. That is the audit join: which policy version governed which write, answered by one query.

Outside every one of them today: prescriptive guidance to an agent (what to verify, which properties to prefer) · trust as a continuum, not permitted / prohibited · audit at the grain of a single mutation.